

Vertical Jump Testing Using an Earable Accelerometer

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ABSTRACT

In recent years, more and more consumer earphones come equipped with inertial measurement units (IMUs). Using such sensors, past research has broadly explored earables for fitness applications, ranging from tracking higher-level workout activities to extracting precise gait-related parameters. Tying in with this research, we initially evaluated how the accelerometer inside earables may be leveraged to conduct vertical jump testing, which is a common standard measure to assess fitness. In a small study, four participants performed seven jump trials each. Using a simple analytical approach, we find that jump height can be calculated at 0.04 m mean absolute error with a median error of 0.02 m.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile devices*.

KEYWORDS

earables, hearables, vertical jump test, accelerometer

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1 INTRODUCTION

In recent years, wearable devices equipped with inertial measurement units (IMUs) have revolutionized the field of sports science and biomechanics. These devices offer the potential to capture and analyze human motion data accurately and continuously, providing valuable insights into various physical parameters [1]. Among these parameters, jump height plays a crucial role in assessing athletic performance, optimizing training programs, and evaluating an athlete's explosive power [7]. Traditionally, jump height estimation has relied on cumbersome and expensive methods such as video-based motion analysis [2] or force plates [4].

In recent years, wearable-based approaches have been explored, however they may be susceptible to trunk rotation artifacts leading to an error of up to 12.7cm to predict jump height [5]. In that regard,

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earables may be advantageous because, compared to the wrist or hips, the head and consequently the ears remain relatively stable during a vertical jump.

Therefore, the goal of this work is to investigate the potential of earable accelerometer data for vertical jump testing. Earables have been employed broadly for fitness-related research because they naturally integrate into the application context, as many enjoy listening to music and podcasts or watching movies while working out [6]. Using earables, past research, for example, has tracked fitness activities on an exercise level [8] but also more granular for example to extract gait-related parameters [3]. We tie in with this research by contributing a new fitness parameter that may be tracked with earables.

This paper introduces a simple analytical approach and initially evaluates the performance of ear-based accelerometer data for jump height measurement. Overall, the system achieves a 0.04 m mean absolute error and a median error of 0.02 m based on 28 jumps from four participants.

2 MATERIALS AND METHODS

In this section, we introduce the measurement equipment and the analytical approach to calculate jump height.

2.1 Measurement Equipment

For this study, we used an iPad Pro 11" (2nd Generation) and AirPods Pro (2nd Generation). The AirPods' IMU consists of 9 DOF. Gravity is removed from the raw acceleration by the provided API. The sample frequency was 25 Hz. For our research, we only used the acceleration data in the jump direction. The IMU data was collected by a custom-developed app running on the previously mentioned iPad. The jump height was measured using AR Body Tracking with Apple's ARKit framework running in the same app. We tracked the position of the hip joint of the participant. The acceleration data was filtered using a lowpass filter. We calculate the relative acceleration from the filtered acceleration data by subtracting the previous value from the current one. The hip position data was not altered.

2.2 Position Calculation

We assume that for each time slice the acceleration is constant. To calculate the height we used the following Equation:

$$h_i = \lambda \cdot \sum_{j=1}^i \frac{1}{2} \cdot a_j \cdot t^2 \quad (1)$$

where h_i is the vertical position at a specific time, specified by i ; a is the dataset of relative accelerations in the vertical axis and j defines the specific acceleration in the dataset; t is the time in which the acceleration took place (0.04 seconds based on the 25 Hz sample

rate of the IMU); λ is a linear correction factor to accommodate systematic errors (see subsection 2.3).

To determine the vertical jump height, we then calculate the difference between the highest point in the height data and the mean of the height data.

2.3 Determining the correction factor λ

To account for the systematic underprediction observed in the initial jump height estimation model, a linear correction factor, denoted as λ , is introduced. The aim is to adjust the calculated jump height and improve alignment with the ground truth obtained through AR Body Tracking.

To determine the optimal value for λ , an iterative search process is employed. Multiple values of λ are tested, and the average prediction error for all datasets is computed for each value. The value of λ resulting in the smallest average prediction error is chosen as the optimal correction factor.

3 EVALUATION

To evaluate the performance of the presented approach, we recruited four participants (3 men, 1 woman). Each participant performed 7 jumps.

3.1 Results

Figure 1 shows the ground truth and the vertical position measured over time of one of the jumps collected in our study. The calculated position is subject to noise but overall, has similar characteristics as the ground truth.

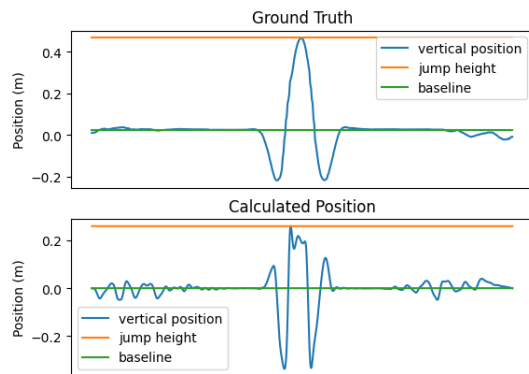


Figure 1: The ground truth and measured vertical position.

Figure 2 summarizes the calculated and measured jump height of each trial. The calculated jump height is corrected by the linear factor λ (see subsection 2.3) to accommodate the systematic underprediction. Overall, this yields a mean absolute error of 0.04 m with a median of 0.02 m across all 28 jumps.

AR as the Ground Truth. In order to evaluate the performance of AR Body Tracking, we used a camera and a meterstick to perform a manual video-based analysis of the data sets to compare the results. We determined that the AR Body tracking had an absolute average error of 0.01m.

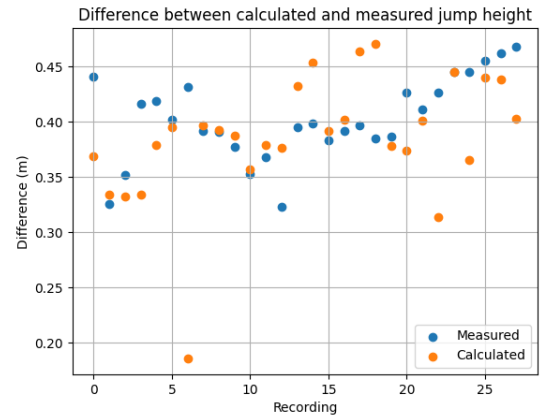


Figure 2: Measured and corrected calculated jump heights of all 28 trials show promising results.

4 DISCUSSION

This study focused on assessing the performance and reliability of earable devices in estimating jump height using accelerometer data, with the ground truth obtained through AR body tracking. The evaluation results indicate that while there was some variability, overall, the earable devices demonstrated good performance and reliability in estimating the jump height.

One of the key aspects evaluated was the accuracy of jump height estimation provided by earable devices. The study found that the average error in jump height estimation using the earable devices was 0.04 meters. This indicates that on average, the earable devices were able to estimate jump height with a relatively small margin of error.

One potential factor contributing to the estimation error is the limited use of vertical acceleration data in the calculations. The study acknowledges that a portion of the acceleration during the jump might have been distributed across the other axes, potentially leading to inaccuracies in jump height estimation. This limitation highlights the need for further exploration of more comprehensive algorithms that take into account the multi-axis acceleration data captured by the earable devices. Future studies could consider incorporating additional sensor fusion techniques to mitigate this issue and improve the accuracy of jump height estimation.

Despite this limitation, the evaluation demonstrated that the earable devices still provided reasonably accurate estimations of jump height. The findings suggest that, even with the focus primarily on vertical acceleration, the earable devices were able to capture the essential motion data necessary for estimating jump height.

5 CONCLUSION

In this paper, we initially investigated how the accelerometer of an earable can be used to estimate the vertical jump height of a user. Our initial study based on 28 trials of four participants revealed a mean absolute error of 0.04 m with a median error of 0.02 m. The presented approach is very simple and only uses one axis of the accelerometer. In future work, the presented results may be expanded by introducing a more advanced processing pipeline, investigating other jump-related fitness tests, and recruiting more participants to increase the reliability of the presented results.

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REFERENCES

- [1] Jacob S Arlotti, William O Carroll, Youness Afifi, Purva Talegaonkar, Luciano Albuquerque, John E Ball, Harish Chander, Adam Petway, et al. 2022. Benefits of IMU-based Wearables in Sports Medicine: Narrative Review. *International Journal of Kinesiology and Sports Science* 10, 1 (2022), 36–43.
- [2] Carlos Balsalobre-Fernández, Carlos M Tejero-González, Juan del Campo-Vecino, and Nicolás Bavaresco. 2014. The concurrent validity and reliability of a low-cost, high-speed camera-based method for measuring the flight time of vertical jumps. *The Journal of Strength & Conditioning Research* 28, 2 (2014), 528–533.
- [3] Delaram Jarchi, Charence Wong, Richard Mark Kwasnicki, Ben Heller, Garry A Tew, and Guang-Zhong Yang. 2014. Gait parameter estimation from a miniaturized ear-worn sensor using singular spectrum analysis and longest common subsequence. *IEEE Transactions on Biomedical Engineering* 61, 4 (2014), 1261–1273.
- [4] Nicholas P Linthorne. 2001. Analysis of standing vertical jumps using a force platform. *American Journal of Physics* 69, 11 (2001), 1198–1204.
- [5] Pietro Picerno, Valentina Camomilla, and Laura Capranica. 2011. Countermovement jump performance assessment using a wearable 3D inertial measurement unit. *Journal of sports sciences* 29, 2 (2011), 139–146.
- [6] Tobias Röddiger, Christopher Clarke, Paula Breitling, Tim Schneegans, Haibin Zhao, Hans Gellersen, and Michael Beigl. 2022. Sensing with Earables: A Systematic Literature Review and Taxonomy of Phenomena. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3 (2022), 1–57.
- [7] Pierre Samozino, Jean-Benoît Morin, Frédérique Hintzy, and Alain Belli. 2008. A simple method for measuring force, velocity and power output during squat jump. *Journal of biomechanics* 41, 14 (2008), 2940–2945.
- [8] David Strömbäck, Sangxia Huang, and Valentin Radu. 2020. Mm-fit: Multimodal deep learning for automatic exercise logging across sensing devices. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–22.